

Appendix

Uniqueness of fit-width solution when $r \geq 1$

Let (a, b) be a minimal fit-width point. We prove there is no b' such that (a, b') is a fit-width point when $r \geq 1$ and (a, b) corresponds to the best solution, meaning that $s_w > s_h$ for (a_h, b_h) minimal fit-width point.

Assume towards a contradiction there exists $b' \neq b$ such that (a, b') is a fit-width point. Based on (14), we can assume without loss of generality that $b = \lceil \frac{n}{a} \rceil$ and that $b' = b + 1$. By (17), the minimality of a and monotonicity of the compared terms with respect to a we have:

$$a - 1 < r \left\lceil \frac{n}{a - 1} \right\rceil \quad (\text{U1})$$

We further know from (14) that $n \leq ab$ and $a \geq r(b + 1)$, so $a - 1 \geq b$ for $r \geq 1$. Let's now bound $\lceil \frac{n}{a-1} \rceil$:

$$\begin{aligned} \frac{n}{a-1} &= \frac{n}{a} \cdot \frac{a}{a-1} \leq \frac{ab}{a} \cdot \left(1 + \frac{1}{a-1}\right) = b + \frac{b}{a-1} \leq b + 1 \\ &\implies \left\lceil \frac{n}{a-1} \right\rceil \leq b + 1 \end{aligned} \quad (\text{U2})$$

Assume that $\lceil \frac{n}{a-1} \rceil \leq b$. By (U1) we have that:

$$a - 1 < rb \leq a \implies a = \lceil rb \rceil$$

But under the assumption that $(a, b + 1)$ is valid:

$$a \geq r(b + 1) \implies \lceil rb \rceil \geq r(b + 1) \geq rb + 1$$

A contradiction which together with (U2) determines $\lceil \frac{n}{a-1} \rceil = b + 1$. It follows using (U1) that:

$$a - 1 < r(b + 1) \wedge n \leq (a - 1)(b + 1)$$

The point $(a - 1, b + 1)$ satisfies (14): it is fit-height (see [graph C](#) for a real example). Since we've assumed b_h is minimal, it must be that $b + 1 \geq b_h$, but this result contradicts the assumption that the fit-width solution is better:

$$s > s_h \implies r(b + 1) \leq a < rb_h \implies b + 1 < b_h$$

In conclusion, no $b' \neq b$ can exist, therefore (a, b) is unique.